

55. What are the truth values of these statements?
- $\exists!xP(x) \rightarrow \exists xP(x)$
 - $\forall xP(x) \rightarrow \exists!xP(x)$
 - $\exists!x\neg P(x) \rightarrow \neg\forall xP(x)$
56. Write out $\exists!xP(x)$, where the domain consists of the integers 1, 2, and 3, in terms of negations, conjunctions, and disjunctions.
57. Given the Prolog facts in Example 28, what would Prolog return given these queries?
- `?instructor(chan, math273)`
 - `?instructor(patel, cs301)`
 - `?enrolled(X, cs301)`
 - `?enrolled(kiko, Y)`
 - `?teaches(grossman, Y)`
58. Given the Prolog facts in Example 28, what would Prolog return when given these queries?
- `?enrolled(kevin, ee222)`
 - `?enrolled(kiko, math273)`
 - `?instructor(grossman, X)`
 - `?instructor(X, cs301)`
 - `?teaches(X, kevin)`
59. Suppose that Prolog facts are used to define the predicates *mother*(*M*, *Y*) and *father*(*F*, *X*), which represent that *M* is the mother of *Y* and *F* is the father of *X*, respectively. Give a Prolog rule to define the predicate *sibling*(*X*, *Y*), which represents that *X* and *Y* are siblings (that is, have the same mother and the same father).
60. Suppose that Prolog facts are used to define the predicates *mother*(*M*, *Y*) and *father*(*F*, *X*), which represent that *M* is the mother of *Y* and *F* is the father of *X*, respectively. Give a Prolog rule to define the predicate *grandfather*(*X*, *Y*), which represents that *X* is the grandfather of *Y*. [Hint: You can write a disjunction in Prolog either by using a semicolon to separate predicates or by putting these predicates on separate lines.]
- Exercises 61–64 are based on questions found in the book *Symbolic Logic* by Lewis Carroll.
61. Let *P*(*x*), *Q*(*x*), and *R*(*x*) be the statements “*x* is a professor,” “*x* is ignorant,” and “*x* is vain,” respectively. Express each of these statements using quantifiers; logical connectives; and *P*(*x*), *Q*(*x*), and *R*(*x*), where the domain consists of all people.
- No professors are ignorant.
 - All ignorant people are vain.
 - No professors are vain.
 - Does (c) follow from (a) and (b)?
62. Let *P*(*x*), *Q*(*x*), and *R*(*x*) be the statements “*x* is a clear explanation,” “*x* is satisfactory,” and “*x* is an excuse,” respectively. Suppose that the domain for *x* consists of all English text. Express each of these statements using quantifiers, logical connectives, and *P*(*x*), *Q*(*x*), and *R*(*x*).
- All clear explanations are satisfactory.
 - Some excuses are unsatisfactory.
 - Some excuses are not clear explanations.
 - Does (c) follow from (a) and (b)?
63. Let *P*(*x*), *Q*(*x*), *R*(*x*), and *S*(*x*) be the statements “*x* is a baby,” “*x* is logical,” “*x* is able to manage a crocodile,” and “*x* is despised,” respectively. Suppose that the domain consists of all people. Express each of these statements using quantifiers; logical connectives; and *P*(*x*), *Q*(*x*), *R*(*x*), and *S*(*x*).
- Babies are illogical.
 - Nobody is despised who can manage a crocodile.
 - Illogical persons are despised.
 - Babies cannot manage crocodiles.
 - Does (d) follow from (a), (b), and (c)? If not, is there a correct conclusion?
64. Let *P*(*x*), *Q*(*x*), *R*(*x*), and *S*(*x*) be the statements “*x* is a duck,” “*x* is one of my poultry,” “*x* is an officer,” and “*x* is willing to waltz,” respectively. Express each of these statements using quantifiers; logical connectives; and *P*(*x*), *Q*(*x*), *R*(*x*), and *S*(*x*).
- No ducks are willing to waltz.
 - No officers ever decline to waltz.
 - All my poultry are ducks.
 - My poultry are not officers.
 - Does (d) follow from (a), (b), and (c)? If not, is there a correct conclusion?

1.5 Nested Quantifiers

1.5.1 Introduction

In Section 1.4 we defined the existential and universal quantifiers and showed how they can be used to represent mathematical statements. We also explained how they can be used to translate English sentences into logical expressions. However, in Section 1.4 we avoided **nested quantifiers**, where one quantifier is within the scope of another, such as

$$\forall x \exists y (x + y = 0).$$

Note that everything within the scope of a quantifier can be thought of as a propositional function. For example,

$$\forall x \exists y (x + y = 0)$$

is the same thing as $\forall x Q(x)$, where $Q(x)$ is $\exists y P(x, y)$, where $P(x, y)$ is $x + y = 0$.

Nested quantifiers commonly occur in mathematics and computer science. Although nested quantifiers can sometimes be difficult to understand, the rules we have already studied in Section 1.4 can help us use them. In this section we will gain experience working with nested quantifiers. We will see how to use nested quantifiers to express mathematical statements such as “The sum of two positive integers is always positive.” We will show how nested quantifiers can be used to translate English sentences such as “Everyone has exactly one best friend” into logical statements. Moreover, we will gain experience working with the negations of statements involving nested quantifiers.

1.5.2 Understanding Statements Involving Nested Quantifiers

To understand statements involving nested quantifiers, we need to unravel what the quantifiers and predicates that appear mean. This is illustrated in Examples 1 and 2.

EXAMPLE 1 Assume that the domain for the variables x and y consists of all real numbers. The statement

$$\forall x \forall y (x + y = y + x)$$

**Extra
Examples** 

says that $x + y = y + x$ for all real numbers x and y . This is the commutative law for addition of real numbers. Likewise, the statement

$$\forall x \exists y (x + y = 0)$$

says that for every real number x there is a real number y such that $x + y = 0$. This states that every real number has an additive inverse. Similarly, the statement

$$\forall x \forall y \forall z (x + (y + z) = (x + y) + z)$$

is the associative law for addition of real numbers. 

EXAMPLE 2 Translate into English the statement

$$\forall x \forall y ((x > 0) \wedge (y < 0) \rightarrow (xy < 0)),$$

where the domain for both variables consists of all real numbers.

Solution: This statement says that for every real number x and for every real number y , if $x > 0$ and $y < 0$, then $xy < 0$. That is, this statement says that for real numbers x and y , if x is positive and y is negative, then xy is negative. This can be stated more succinctly as “The product of a positive real number and a negative real number is always a negative real number.” 

THINKING OF QUANTIFICATION AS LOOPS In working with quantifications of more than one variable, it is sometimes helpful to think in terms of nested loops. (If there are infinitely many elements in the domain of some variable, we cannot actually loop through all values. Nevertheless, this way of thinking is helpful in understanding nested quantifiers.) For example, to see whether $\forall x \forall y P(x, y)$ is true, we loop through the values for x , and for each x we loop through the values for y . If we find that for all values of x that $P(x, y)$ is true for all values of y ,

we have determined that $\forall x \forall y P(x, y)$ is true. If we ever hit a value x for which we hit a value y for which $P(x, y)$ is false, we have shown that $\forall x \forall y P(x, y)$ is false.

Similarly, to determine whether $\forall x \exists y P(x, y)$ is true, we loop through the values for x . For each x we loop through the values for y until we find a y for which $P(x, y)$ is true. If for every x we hit such a y , then $\forall x \exists y P(x, y)$ is true; if for some x we never hit such a y , then $\forall x \exists y P(x, y)$ is false.

To see whether $\exists x \forall y P(x, y)$ is true, we loop through the values for x until we find an x for which $P(x, y)$ is always true when we loop through all values for y . Once we find such an x , we know that $\exists x \forall y P(x, y)$ is true. If we never hit such an x , then we know that $\exists x \forall y P(x, y)$ is false.

Finally, to see whether $\exists x \exists y P(x, y)$ is true, we loop through the values for x , where for each x we loop through the values for y until we hit an x for which we hit a y for which $P(x, y)$ is true. The statement $\exists x \exists y P(x, y)$ is false only if we never hit an x for which we hit a y such that $P(x, y)$ is true.

1.5.3 The Order of Quantifiers

Many mathematical statements involve multiple quantifications of propositional functions involving more than one variable. It is important to note that the order of the quantifiers is important, unless all the quantifiers are universal quantifiers or all are existential quantifiers.

These remarks are illustrated by Examples 3–5.

EXAMPLE 3 Let $P(x, y)$ be the statement “ $x + y = y + x$.” What are the truth values of the quantifications $\forall x \forall y P(x, y)$ and $\forall y \forall x P(x, y)$, where the domain for all variables consists of all real numbers?

Extra Examples ➤

Solution: The quantification

$$\forall x \forall y P(x, y)$$

denotes the proposition

“For all real numbers x , for all real numbers y , $x + y = y + x$.”

Because $P(x, y)$ is true for all real numbers x and y (it is the commutative law for addition, which is an axiom for the real numbers—see Appendix 1), the proposition $\forall x \forall y P(x, y)$ is true. Note that the statement $\forall y \forall x P(x, y)$ says “For all real numbers y , for all real numbers x , $x + y = y + x$.” This has the same meaning as the statement “For all real numbers x , for all real numbers y , $x + y = y + x$.” That is, $\forall x \forall y P(x, y)$ and $\forall y \forall x P(x, y)$ have the same meaning, and both are true. This illustrates the principle that the order of nested universal quantifiers in a statement without other quantifiers can be changed without changing the meaning of the quantified statement. ◀

EXAMPLE 4 Let $Q(x, y)$ denote “ $x + y = 0$.” What are the truth values of the quantifications $\exists y \forall x Q(x, y)$ and $\forall x \exists y Q(x, y)$, where the domain for all variables consists of all real numbers?

Solution: The quantification

$$\exists y \forall x Q(x, y)$$

denotes the proposition

“There is a real number y such that for every real number x , $Q(x, y)$.”

No matter what value of y is chosen, there is only one value of x for which $x + y = 0$. Because there is no real number y such that $x + y = 0$ for all real numbers x , the statement $\exists y \forall x Q(x, y)$ is false.

The quantification

$$\forall x \exists y Q(x, y)$$

denotes the proposition

“For every real number x there is a real number y such that $Q(x, y)$.”

Given a real number x , there is a real number y such that $x + y = 0$; namely, $y = -x$. Hence, the statement $\forall x \exists y Q(x, y)$ is true. 

Be careful with the order of existential and universal quantifiers!

Example 4 illustrates that the order in which quantifiers appear makes a difference. The statements $\exists y \forall x P(x, y)$ and $\forall x \exists y P(x, y)$ are not logically equivalent. The statement $\exists y \forall x P(x, y)$ is true if and only if there is a y that makes $P(x, y)$ true for every x . So, for this statement to be true, there must be a particular value of y for which $P(x, y)$ is true regardless of the choice of x . On the other hand, $\forall x \exists y P(x, y)$ is true if and only if for every value of x there is a value of y for which $P(x, y)$ is true. So, for this statement to be true, no matter which x you choose, there must be a value of y (possibly depending on the x you choose) for which $P(x, y)$ is true. In other words, in the second case, y can depend on x , whereas in the first case, y is a constant independent of x .

From these observations, it follows that if $\exists y \forall x P(x, y)$ is true, then $\forall x \exists y P(x, y)$ must also be true. However, if $\forall x \exists y P(x, y)$ is true, it is not necessary for $\exists y \forall x P(x, y)$ to be true. (See Supplementary Exercises 30 and 31.)

Table 1 summarizes the meanings of the different possible quantifications involving two variables.

Quantifications of more than two variables are also common, as Example 5 illustrates.

EXAMPLE 5 Let $Q(x, y, z)$ be the statement “ $x + y = z$.” What are the truth values of the statements $\forall x \forall y \exists z Q(x, y, z)$ and $\exists z \forall x \forall y Q(x, y, z)$, where the domain of all variables consists of all real numbers?

Solution: Suppose that x and y are assigned values. Then, there exists a real number z such that $x + y = z$. Consequently, the quantification

$$\forall x \forall y \exists z Q(x, y, z),$$

which is the statement

“For all real numbers x and for all real numbers y there is a real number z such that $x + y = z$,”

TABLE 1 Quantifications of Two Variables.

Statement	When True?	When False?
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y .

is true. The order of the quantification here is important, because the quantification

$$\exists z \forall x \forall y Q(x, y, z),$$

which is the statement

“There is a real number z such that for all real numbers x and for all real numbers y it is true that $x + y = z$,”

is false, because there is no value of z that satisfies the equation $x + y = z$ for all values of x and y . 

1.5.4 Translating Mathematical Statements into Statements Involving Nested Quantifiers

Mathematical statements expressed in English can be translated into logical expressions, as Examples 6–8 show.

EXAMPLE 6 Translate the statement “The sum of two positive integers is always positive” into a logical expression.

Extra Examples 

Solution: To translate this statement into a logical expression, we first rewrite it so that the implied quantifiers and a domain are shown: “For every two integers, if these integers are both positive, then the sum of these integers is positive.” Next, we introduce the variables x and y to obtain “For all positive integers x and y , $x + y$ is positive.” Consequently, we can express this statement as

$$\forall x \forall y ((x > 0) \wedge (y > 0) \rightarrow (x + y > 0)),$$

where the domain for both variables consists of all integers. Note that we could also translate this using the positive integers as the domain. Then the statement “The sum of two positive integers is always positive” becomes “For every two positive integers, the sum of these integers is positive.” We can express this as

$$\forall x \forall y (x + y > 0),$$

where the domain for both variables consists of all positive integers. 

EXAMPLE 7 Translate the statement “Every real number except zero has a multiplicative inverse.” (A **multiplicative inverse** of a real number x is a real number y such that $xy = 1$.)

Solution: We first rewrite this as “For every real number x except zero, x has a multiplicative inverse.” We can rewrite this as “For every real number x , if $x \neq 0$, then there exists a real number y such that $xy = 1$.” This can be rewritten as

$$\forall x ((x \neq 0) \rightarrow \exists y (xy = 1)).$$


One example that you may be familiar with is the concept of limit, which is important in calculus.

EXAMPLE 8 (*Requires calculus*) Use quantifiers to express the definition of the limit of a real-valued function $f(x)$ of a real variable x at a point a in its domain.

Solution: Recall that the definition of the statement

$$\lim_{x \rightarrow a} f(x) = L$$

is: For every real number $\epsilon > 0$ there exists a real number $\delta > 0$ such that $|f(x) - L| < \epsilon$ whenever $0 < |x - a| < \delta$. This definition of a limit can be phrased in terms of quantifiers by

$$\forall \epsilon \exists \delta \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon),$$

where the domain for the variables δ and ϵ consists of all positive real numbers and for x consists of all real numbers.

This definition can also be expressed as

$$\forall \epsilon > 0 \exists \delta > 0 \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon)$$

when the domain for the variables ϵ and δ consists of all real numbers, rather than just the positive real numbers. [Here, restricted quantifiers have been used. Recall that $\forall x > 0 P(x)$ means that for all x with $x > 0$, $P(x)$ is true.] ◀

1.5.5 Translating from Nested Quantifiers into English

Expressions with nested quantifiers expressing statements in English can be quite complicated. The first step in translating such an expression is to write out what the quantifiers and predicates in the expression mean. The next step is to express this meaning in a simpler sentence. This process is illustrated in Examples 9 and 10.

EXAMPLE 9 Translate the statement

$$\forall x (C(x) \vee \exists y (C(y) \wedge F(x, y)))$$

into English, where $C(x)$ is “ x has a computer,” $F(x, y)$ is “ x and y are friends,” and the domain for both x and y consists of all students in your school.

Solution: The statement says that for every student x in your school, x has a computer or there is a student y such that y has a computer and x and y are friends. In other words, every student in your school has a computer or has a friend who has a computer. ◀

EXAMPLE 10 Translate the statement

$$\exists x \forall y \forall z ((F(x, y) \wedge F(x, z) \wedge (y \neq z)) \rightarrow \neg F(y, z))$$

into English, where $F(a, b)$ means a and b are friends and the domain for x , y , and z consists of all students in your school.

Solution: We first examine the expression $(F(x, y) \wedge F(x, z) \wedge (y \neq z)) \rightarrow \neg F(y, z)$. This expression says that if students x and y are friends, and students x and z are friends, and furthermore, if y and z are not the same student, then y and z are not friends. It follows that the original statement, which is triply quantified, says that there is a student x such that for all students y and all students z other than y , if x and y are friends and x and z are friends, then y and z are not friends. In other words, there is a student none of whose friends are also friends with each other. ◀

1.5.6 Translating English Sentences into Logical Expressions

In Section 1.4 we showed how quantifiers can be used to translate sentences into logical expressions. However, we avoided sentences whose translation into logical expressions required the use of nested quantifiers. We now address the translation of such sentences.

EXAMPLE 11 Express the statement “If a person is female and is a parent, then this person is someone’s mother” as a logical expression involving predicates, quantifiers with a domain consisting of all people, and logical connectives.

Solution: The statement “If a person is female and is a parent, then this person is someone’s mother” can be expressed as “For every person x , if person x is female and person x is a parent, then there exists a person y such that person x is the mother of person y .” We introduce the propositional functions $F(x)$ to represent “ x is female,” $P(x)$ to represent “ x is a parent,” and $M(x, y)$ to represent “ x is the mother of y .” The original statement can be represented as

$$\forall x((F(x) \wedge P(x)) \rightarrow \exists y M(x, y)).$$

Using the null quantification rule in part (b) of Exercise 49 in Section 1.4, we can move $\exists y$ to the left so that it appears just after $\forall x$, because y does not appear in $F(x) \wedge P(x)$. We obtain the logically equivalent expression

$$\forall x \exists y ((F(x) \wedge P(x)) \rightarrow M(x, y)).$$

EXAMPLE 12 Express the statement “Everyone has exactly one best friend” as a logical expression involving predicates, quantifiers with a domain consisting of all people, and logical connectives.

Solution: The statement “Everyone has exactly one best friend” can be expressed as “For every person x , person x has exactly one best friend.” Introducing the universal quantifier, we see that this statement is the same as “ $\forall x$ (person x has exactly one best friend),” where the domain consists of all people.

To say that x has exactly one best friend means that there is a person y who is the best friend of x , and furthermore, that for every person z , if person z is not person y , then z is not the best friend of x . When we introduce the predicate $B(x, y)$ to be the statement “ y is the best friend of x ,” the statement that x has exactly one best friend can be represented as

$$\exists y(B(x, y) \wedge \forall z((z \neq y) \rightarrow \neg B(x, z))).$$

Consequently, our original statement can be expressed as

$$\forall x \exists y (B(x, y) \wedge \forall z((z \neq y) \rightarrow \neg B(x, z))).$$

[Note that we can write this statement as $\forall x \exists! y B(x, y)$, where $\exists!$ is the “uniqueness quantifier” defined in Section 1.4.]

EXAMPLE 13 Use quantifiers to express the statement “There is a woman who has taken a flight on every airline in the world.”

Solution: Let $P(w, f)$ be “ w has taken f ” and $Q(f, a)$ be “ f is a flight on a .” We can express the statement as

$$\exists w \forall a \exists f (P(w, f) \wedge Q(f, a)),$$

where the domains of discourse for w, f , and a consist of all the women in the world, all airplane flights, and all airlines, respectively.

The statement could also be expressed as

$$\exists w \forall a \exists f R(w, f, a),$$

where $R(w, f, a)$ is “ w has taken f on a .” Although this is more compact, it somewhat obscures the relationships among the variables. Consequently, the first solution is usually preferable. 

1.5.7 Negating Nested Quantifiers

Assessment

Statements involving nested quantifiers can be negated by successively applying the rules for negating statements involving a single quantifier. This is illustrated in Examples 14–16.

EXAMPLE 14

Extra Examples

Express the negation of the statement $\forall x \exists y (xy = 1)$ so that no negation precedes a quantifier.

Solution: By successively applying De Morgan’s laws for quantifiers in Table 2 of Section 1.4, we can move the negation in $\neg \forall x \exists y (xy = 1)$ inside all the quantifiers. We find that $\neg \forall x \exists y (xy = 1)$ is equivalent to $\exists x \neg \exists y (xy = 1)$, which is equivalent to $\exists x \forall y \neg (xy = 1)$. Because $\neg (xy = 1)$ can be expressed more simply as $xy \neq 1$, we conclude that our negated statement can be expressed as $\exists x \forall y (xy \neq 1)$. 

EXAMPLE 15

Use quantifiers to express the statement that “There does not exist a woman who has taken a flight on every airline in the world.”

Solution: This statement is the negation of the statement “There is a woman who has taken a flight on every airline in the world” from Example 13. By Example 13, our statement can be expressed as $\neg \exists w \forall a \exists f (P(w, f) \wedge Q(f, a))$, where $P(w, f)$ is “ w has taken f ” and $Q(f, a)$ is “ f is a flight on a .” By successively applying De Morgan’s laws for quantifiers in Table 2 of Section 1.4 to move the negation inside successive quantifiers and by applying De Morgan’s law for negating a conjunction in the last step, we find that our statement is equivalent to each of this sequence of statements:

$$\begin{aligned} \neg \exists w \neg \forall a \neg \exists f (P(w, f) \wedge Q(f, a)) &\equiv \forall w \exists a \neg \exists f (P(w, f) \wedge Q(f, a)) \\ &\equiv \forall w \exists a \forall f \neg (P(w, f) \wedge Q(f, a)) \\ &\equiv \forall w \exists a \forall f (\neg P(w, f) \vee \neg Q(f, a)). \end{aligned}$$

This last statement states “For every woman there is an airline such that for all flights, this woman has not taken that flight or that flight is not on this airline.” 

EXAMPLE 16

(Requires calculus) Use quantifiers and predicates to express the fact that $\lim_{x \rightarrow a} f(x)$ does not exist where $f(x)$ is a real-valued function of a real variable x and a belongs to the domain of f

Solution: To say that $\lim_{x \rightarrow a} f(x)$ does not exist means that for all real numbers L , $\lim_{x \rightarrow a} f(x) \neq L$. By using Example 8, the statement $\lim_{x \rightarrow a} f(x) \neq L$ can be expressed as

$$\neg \forall \epsilon > 0 \exists \delta > 0 \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon).$$

Successively applying the rules for negating quantified expressions, we construct this sequence of equivalent statements:

$$\begin{aligned}
 & \neg \forall \epsilon > 0 \exists \delta > 0 \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon) \\
 & \equiv \exists \epsilon > 0 \neg \exists \delta > 0 \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon) \\
 & \equiv \exists \epsilon > 0 \forall \delta > 0 \neg \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon) \\
 & \equiv \exists \epsilon > 0 \forall \delta > 0 \exists x \neg (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon) \\
 & \equiv \exists \epsilon > 0 \forall \delta > 0 \exists x (0 < |x - a| < \delta \wedge |f(x) - L| \geq \epsilon).
 \end{aligned}$$

In the last step we used the equivalence $\neg(p \rightarrow q) \equiv p \wedge \neg q$, which follows from the fifth equivalence in Table 7 of Section 1.3.

Because the statement “ $\lim_{x \rightarrow a} f(x)$ does not exist” means for all real numbers L , $\lim_{x \rightarrow a} f(x) \neq L$, this can be expressed as

$$\forall L \exists \epsilon > 0 \forall \delta > 0 \exists x (0 < |x - a| < \delta \wedge |f(x) - L| \geq \epsilon).$$

This last statement says that for every real number L there is a real number $\epsilon > 0$ such that for every real number $\delta > 0$, there exists a real number x such that $0 < |x - a| < \delta$ and $|f(x) - L| \geq \epsilon$. ◀

Exercises

- Translate these statements into English, where the domain for each variable consists of all real numbers.
 - $\forall x \exists y (x < y)$
 - $\forall x \forall y (((x \geq 0) \wedge (y \geq 0)) \rightarrow (xy \geq 0))$
 - $\forall x \forall y \exists z (xy = z)$
- Translate these statements into English, where the domain for each variable consists of all real numbers.
 - $\exists x \forall y (xy = y)$
 - $\forall x \forall y (((x \geq 0) \wedge (y < 0)) \rightarrow (x - y > 0))$
 - $\forall x \forall y \exists z (x = y + z)$
- Let $Q(x, y)$ be the statement “ x has sent an e-mail message to y ,” where the domain for both x and y consists of all students in your class. Express each of these quantifications in English.

a) $\exists x \exists y Q(x, y)$	b) $\exists x \forall y Q(x, y)$
c) $\forall x \exists y Q(x, y)$	d) $\exists y \forall x Q(x, y)$
e) $\forall y \exists x Q(x, y)$	f) $\forall x \forall y Q(x, y)$
- Let $P(x, y)$ be the statement “Student x has taken class y ,” where the domain for x consists of all students in your class and for y consists of all computer science courses at your school. Express each of these quantifications in English.

a) $\exists x \exists y P(x, y)$	b) $\exists x \forall y P(x, y)$
c) $\forall x \exists y P(x, y)$	d) $\exists y \forall x P(x, y)$
e) $\forall y \exists x P(x, y)$	f) $\forall x \forall y P(x, y)$
- Let $W(x, y)$ mean that student x has visited website y , where the domain for x consists of all students in your school and the domain for y consists of all websites. Express each of these statements by a simple English sentence.
 - $W(\text{Sarah Smith}, \text{www.att.com})$
 - $\exists x W(x, \text{www.imdb.org})$
 - $\exists y W(\text{José Orez}, y)$
 - $\exists y (W(\text{Ashok Puri}, y) \wedge W(\text{Cindy Yoon}, y))$
 - $\exists y \forall z (y \neq (\text{David Belcher}) \wedge (W(\text{David Belcher}, z) \rightarrow W(y, z)))$
 - $\exists x \exists y \forall z ((x \neq y) \wedge (W(x, z) \leftrightarrow W(y, z)))$
- Let $C(x, y)$ mean that student x is enrolled in class y , where the domain for x consists of all students in your school and the domain for y consists of all classes being given at your school. Express each of these statements by a simple English sentence.
 - $C(\text{Randy Goldberg}, \text{CS 252})$
 - $\exists x C(x, \text{Math 695})$
 - $\exists y C(\text{Carol Sitea}, y)$
 - $\exists x (C(x, \text{Math 222}) \wedge C(x, \text{CS 252}))$
 - $\exists x \exists y \forall z ((x \neq y) \wedge (C(x, z) \rightarrow C(y, z)))$
 - $\exists x \exists y \forall z ((x \neq y) \wedge (C(x, z) \leftrightarrow C(y, z)))$
- Let $T(x, y)$ mean that student x likes cuisine y , where the domain for x consists of all students at your school and the domain for y consists of all cuisines. Express each of these statements by a simple English sentence.
 - $\neg T(\text{Abdallah Hussein}, \text{Japanese})$

- b) $\exists x T(x, \text{Korean}) \wedge \forall x T(x, \text{Mexican})$
 c) $\exists y (T(\text{Monique Arsenault}, y) \vee T(\text{Jay Johnson}, y))$
 d) $\forall x \forall z \exists y ((x \neq z) \rightarrow \neg(T(x, y) \wedge T(z, y)))$
 e) $\exists x \exists z \forall y (T(x, y) \leftrightarrow T(z, y))$
 f) $\forall x \forall z \exists y (T(x, y) \leftrightarrow T(z, y))$
8. Let $Q(x, y)$ be the statement “Student x has been a contestant on quiz show y .” Express each of these sentences in terms of $Q(x, y)$, quantifiers, and logical connectives, where the domain for x consists of all students at your school and for y consists of all quiz shows on television.
- There is a student at your school who has been a contestant on a television quiz show.
 - No student at your school has ever been a contestant on a television quiz show.
 - There is a student at your school who has been a contestant on *Jeopardy!* and on *Wheel of Fortune*.
 - Every television quiz show has had a student from your school as a contestant.
 - At least two students from your school have been contestants on *Jeopardy!*.
9. Let $L(x, y)$ be the statement “ x loves y ,” where the domain for both x and y consists of all people in the world. Use quantifiers to express each of these statements.
- Everybody loves Jerry.
 - Everybody loves somebody.
 - There is somebody whom everybody loves.
 - Nobody loves everybody.
 - There is somebody whom Lydia does not love.
 - There is somebody whom no one loves.
 - There is exactly one person whom everybody loves.
 - There are exactly two people whom Lynn loves.
 - Everyone loves himself or herself.
 - There is someone who loves no one besides himself or herself.
10. Let $F(x, y)$ be the statement “ x can fool y ,” where the domain consists of all people in the world. Use quantifiers to express each of these statements.
- Everybody can fool Fred.
 - Evelyn can fool everybody.
 - Everybody can fool somebody.
 - There is no one who can fool everybody.
 - Everyone can be fooled by somebody.
 - No one can fool both Fred and Jerry.
 - Nancy can fool exactly two people.
 - There is exactly one person whom everybody can fool.
 - No one can fool himself or herself.
 - There is someone who can fool exactly one person besides himself or herself.
11. Let $S(x)$ be the predicate “ x is a student,” $F(x)$ the predicate “ x is a faculty member,” and $A(x, y)$ the predicate “ x has asked y a question,” where the domain consists of all people associated with your school. Use quantifiers to express each of these statements.
- Lois has asked Professor Michaels a question.
 - Every student has asked Professor Gross a question.
 - Every faculty member has either asked Professor Miller a question or been asked a question by Professor Miller.
 - Some student has not asked any faculty member a question.
 - There is a faculty member who has never been asked a question by a student.
 - Some student has asked every faculty member a question.
 - There is a faculty member who has asked every other faculty member a question.
 - Some student has never been asked a question by a faculty member.
12. Let $I(x)$ be the statement “ x has an Internet connection” and $C(x, y)$ be the statement “ x and y have chatted over the Internet,” where the domain for the variables x and y consists of all students in your class. Use quantifiers to express each of these statements.
- Jerry does not have an Internet connection.
 - Rachel has not chatted over the Internet with Chelsea.
 - Jan and Sharon have never chatted over the Internet.
 - No one in the class has chatted with Bob.
 - Sanjay has chatted with everyone except Joseph.
 - Someone in your class does not have an Internet connection.
 - Not everyone in your class has an Internet connection.
 - Exactly one student in your class has an Internet connection.
 - Everyone except one student in your class has an Internet connection.
 - Everyone in your class with an Internet connection has chatted over the Internet with at least one other student in your class.
 - Someone in your class has an Internet connection but has not chatted with anyone else in your class.
 - There are two students in your class who have not chatted with each other over the Internet.
 - There is a student in your class who has chatted with everyone in your class over the Internet.
 - There are at least two students in your class who have not chatted with the same person in your class.
 - There are two students in the class who between them have chatted with everyone else in the class.
13. Let $M(x, y)$ be “ x has sent y an e-mail message” and $T(x, y)$ be “ x has telephoned y ,” where the domain consists of all students in your class. Use quantifiers to express each of these statements. (Assume that all e-mail messages that were sent are received, which is not the way things often work.)
- Chou has never sent an e-mail message to Koko.
 - Arlene has never sent an e-mail message to or telephoned Sarah.
 - José has never received an e-mail message from Deborah.
 - Every student in your class has sent an e-mail message to Ken.
 - No one in your class has telephoned Nina.

- f) Everyone in your class has either telephoned Avi or sent him an e-mail message.
 - g) There is a student in your class who has sent everyone else in your class an e-mail message.
 - h) There is someone in your class who has either sent an e-mail message or telephoned everyone else in your class.
 - i) There are two different students in your class who have sent each other e-mail messages.
 - j) There is a student who has sent himself or herself an e-mail message.
 - k) There is a student in your class who has not received an e-mail message from anyone else in the class and who has not been called by any other student in the class.
 - l) Every student in the class has either received an e-mail message or received a telephone call from another student in the class.
 - m) There are at least two students in your class such that one student has sent the other e-mail and the second student has telephoned the first student.
 - n) There are two different students in your class who between them have sent an e-mail message to or telephoned everyone else in the class.
14. Use quantifiers and predicates with more than one variable to express these statements.
- a) There is a student in this class who can speak Hindi.
 - b) Every student in this class plays some sport.
 - c) Some student in this class has visited Alaska but has not visited Hawaii.
 - d) All students in this class have learned at least one programming language.
 - e) There is a student in this class who has taken every course offered by one of the departments in this school.
 - f) Some student in this class grew up in the same town as exactly one other student in this class.
 - g) Every student in this class has chatted with at least one other student in at least one chat group.
15. Use quantifiers and predicates with more than one variable to express these statements.
- a) Every computer science student needs a course in discrete mathematics.
 - b) There is a student in this class who owns a personal computer.
 - c) Every student in this class has taken at least one computer science course.
 - d) There is a student in this class who has taken at least one course in computer science.
 - e) Every student in this class has been in every building on campus.
 - f) There is a student in this class who has been in every room of at least one building on campus.
 - g) Every student in this class has been in at least one room of every building on campus.
16. A discrete mathematics class contains 1 mathematics major who is a freshman, 12 mathematics majors who are sophomores, 15 computer science majors who are sophomores, 2 mathematics majors who are juniors, 2 computer science majors who are juniors, and 1 computer science major who is a senior. Express each of these statements in terms of quantifiers and then determine its truth value.
- a) There is a student in the class who is a junior.
 - b) Every student in the class is a computer science major.
 - c) There is a student in the class who is neither a mathematics major nor a junior.
 - d) Every student in the class is either a sophomore or a computer science major.
 - e) There is a major such that there is a student in the class in every year of study with that major.
17. Express each of these system specifications using predicates, quantifiers, and logical connectives, if necessary.
- a) Every user has access to exactly one mailbox.
 - b) There is a process that continues to run during all error conditions only if the kernel is working correctly.
 - c) All users on the campus network can access all websites whose url has a .edu extension.
 - * d) There are exactly two systems that monitor every remote server.
18. Express each of these system specifications using predicates, quantifiers, and logical connectives, if necessary.
- a) At least one console must be accessible during every fault condition.
 - b) The e-mail address of every user can be retrieved whenever the archive contains at least one message sent by every user on the system.
 - c) For every security breach there is at least one mechanism that can detect that breach if and only if there is a process that has not been compromised.
 - d) There are at least two paths connecting every two distinct endpoints on the network.
 - e) No one knows the password of every user on the system except for the system administrator, who knows all passwords.
19. Express each of these statements using mathematical and logical operators, predicates, and quantifiers, where the domain consists of all integers.
- a) The sum of two negative integers is negative.
 - b) The difference of two positive integers is not necessarily positive.
 - c) The sum of the squares of two integers is greater than or equal to the square of their sum.
 - d) The absolute value of the product of two integers is the product of their absolute values.
20. Express each of these statements using predicates, quantifiers, logical connectives, and mathematical operators where the domain consists of all integers.
- a) The product of two negative integers is positive.
 - b) The average of two positive integers is positive.

- c) The difference of two negative integers is not necessarily negative.
 d) The absolute value of the sum of two integers does not exceed the sum of the absolute values of these integers.
21. Use predicates, quantifiers, logical connectives, and mathematical operators to express the statement that every positive integer is the sum of the squares of four integers.
22. Use predicates, quantifiers, logical connectives, and mathematical operators to express the statement that there is a positive integer that is not the sum of three squares.
23. Express each of these mathematical statements using predicates, quantifiers, logical connectives, and mathematical operators.
- a) The product of two negative real numbers is positive.
 b) The difference of a real number and itself is zero.
 c) Every positive real number has exactly two square roots.
 d) A negative real number does not have a square root that is a real number.
24. Translate each of these nested quantifications into an English statement that expresses a mathematical fact. The domain in each case consists of all real numbers.
- a) $\exists x \forall y (x + y = y)$
 b) $\forall x \forall y ((x \geq 0) \wedge (y < 0)) \rightarrow (x - y > 0)$
 c) $\exists x \exists y ((x \leq 0) \wedge (y \leq 0) \wedge (x - y > 0))$
 d) $\forall x \forall y ((x \neq 0) \wedge (y \neq 0) \leftrightarrow (xy \neq 0))$
25. Translate each of these nested quantifications into an English statement that expresses a mathematical fact. The domain in each case consists of all real numbers.
- a) $\exists x \forall y (xy = y)$
 b) $\forall x \forall y ((x < 0) \wedge (y < 0)) \rightarrow (xy > 0)$
 c) $\exists x \exists y ((x^2 > y) \wedge (x < y))$
 d) $\forall x \forall y \exists z (x + y = z)$
26. Let $Q(x, y)$ be the statement “ $x + y = x - y$.” If the domain for both variables consists of all integers, what are the truth values?
- a) $Q(1, 1)$ b) $Q(2, 0)$
 c) $\forall y Q(1, y)$ d) $\exists x Q(x, 2)$
 e) $\exists x \exists y Q(x, y)$ f) $\forall x \exists y Q(x, y)$
 g) $\exists y \forall x Q(x, y)$ h) $\forall y \exists x Q(x, y)$
 i) $\forall x \forall y Q(x, y)$
27. Determine the truth value of each of these statements if the domain for all variables consists of all integers.
- a) $\forall n \exists m (n^2 < m)$ b) $\exists n \forall m (n < m^2)$
 c) $\forall n \exists m (n + m = 0)$ d) $\exists n \forall m (nm = m)$
 e) $\exists n \exists m (n^2 + m^2 = 5)$ f) $\exists n \exists m (n^2 + m^2 = 6)$
 g) $\exists n \exists m (n + m = 4 \wedge n - m = 1)$
 h) $\exists n \exists m (n + m = 4 \wedge n - m = 2)$
 i) $\forall n \forall m \exists p (p = (m + n)/2)$
28. Determine the truth value of each of these statements if the domain of each variable consists of all real numbers.
- a) $\forall x \exists y (x^2 = y)$ b) $\forall x \exists y (x = y^2)$
 c) $\exists x \forall y (xy = 0)$ d) $\exists x \exists y (x + y \neq y + x)$
- e) $\forall x (x \neq 0 \rightarrow \exists y (xy = 1))$
 f) $\exists x \forall y (y \neq 0 \rightarrow xy = 1)$
 g) $\forall x \exists y (x + y = 1)$
 h) $\exists x \exists y (x + 2y = 2 \wedge 2x + 4y = 5)$
 i) $\forall x \exists y (x + y = 2 \wedge 2x - y = 1)$
 j) $\forall x \forall y \exists z (z = (x + y)/2)$
29. Suppose the domain of the propositional function $P(x, y)$ consists of pairs x and y , where x is 1, 2, or 3 and y is 1, 2, or 3. Write out these propositions using disjunctions and conjunctions.
- a) $\forall x \forall y P(x, y)$ b) $\exists x \exists y P(x, y)$
 c) $\exists x \forall y P(x, y)$ d) $\forall y \exists x P(x, y)$
30. Rewrite each of these statements so that negations appear only within predicates (that is, so that no negation is outside a quantifier or an expression involving logical connectives).
- a) $\neg \exists y \exists x P(x, y)$ b) $\neg \forall x \exists y P(x, y)$
 c) $\neg \exists y (Q(y) \wedge \forall x \neg R(x, y))$
 d) $\neg \exists y (\exists x R(x, y) \vee \forall x S(x, y))$
 e) $\neg \exists y (\forall x \exists z T(x, y, z) \vee \exists x \forall z U(x, y, z))$
31. Express the negations of each of these statements so that all negation symbols immediately precede predicates.
- a) $\forall x \exists y \forall z T(x, y, z)$
 b) $\forall x \exists y P(x, y) \vee \forall x \exists y Q(x, y)$
 c) $\forall x \exists y (P(x, y) \wedge \exists z R(x, y, z))$
 d) $\forall x \exists y (P(x, y) \rightarrow Q(x, y))$
32. Express the negations of each of these statements so that all negation symbols immediately precede predicates.
- a) $\exists z \forall y \forall x T(x, y, z)$
 b) $\exists x \exists y P(x, y) \wedge \forall x \forall y Q(x, y)$
 c) $\exists x \exists y (Q(x, y) \leftrightarrow Q(y, x))$
 d) $\forall y \exists x \exists z (T(x, y, z) \vee Q(x, y))$
33. Rewrite each of these statements so that negations appear only within predicates (that is, so that no negation is outside a quantifier or an expression involving logical connectives).
- a) $\neg \forall x \forall y P(x, y)$ b) $\neg \forall y \exists x P(x, y)$
 c) $\neg \forall y \forall x (P(x, y) \vee Q(x, y))$
 d) $\neg (\exists x \exists y \neg P(x, y) \wedge \forall x \forall y Q(x, y))$
 e) $\neg \forall x (\exists y \forall z P(x, y, z) \wedge \exists z \forall y P(x, y, z))$
34. Find a common domain for the variables x, y , and z for which the statement $\forall x \forall y ((x \neq y) \rightarrow \forall z ((z = x) \vee (z = y)))$ is true and another domain for which it is false.
35. Find a common domain for the variables x, y, z , and w for which the statement $\forall x \forall y \forall z \exists w ((w \neq x) \wedge (w \neq y) \wedge (w \neq z))$ is true and another common domain for these variables for which it is false.
36. Express each of these statements using quantifiers. Then form the negation of the statement so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not simply use the phrase “It is not the case that.”)
- a) No one has lost more than one thousand dollars playing the lottery.
 b) There is a student in this class who has chatted with exactly one other student.

- c) No student in this class has sent e-mail to exactly two other students in this class.
 - d) Some student has solved every exercise in this book.
 - e) No student has solved at least one exercise in every section of this book.
37. Express each of these statements using quantifiers. Then form the negation of the statement so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not simply use the phrase “It is not the case that.”)
- a) Every student in this class has taken exactly two mathematics classes at this school.
 - b) Someone has visited every country in the world except Libya.
 - c) No one has climbed every mountain in the Himalayas.
 - d) Every movie actor has either been in a movie with Kevin Bacon or has been in a movie with someone who has been in a movie with Kevin Bacon.
38. Express the negations of these propositions using quantifiers, and in English.
- a) Every student in this class likes mathematics.
 - b) There is a student in this class who has never seen a computer.
 - c) There is a student in this class who has taken every mathematics course offered at this school.
 - d) There is a student in this class who has been in at least one room of every building on campus.
39. Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all integers.
- a) $\forall x \forall y (x^2 = y^2 \rightarrow x = y)$
 - b) $\forall x \exists y (y^2 = x)$
 - c) $\forall x \forall y (xy \geq x)$
40. Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all integers.
- a) $\forall x \exists y (x = 1/y)$
 - b) $\forall x \exists y (y^2 - x < 100)$
 - c) $\forall x \forall y (x^2 \neq y^3)$
41. Use quantifiers to express the associative law for multiplication of real numbers.
42. Use quantifiers to express the distributive laws of multiplication over addition for real numbers.
43. Use quantifiers and logical connectives to express the fact that every linear polynomial (that is, polynomial of degree 1) with real coefficients and where the coefficient of x is nonzero, has exactly one real root.
44. Use quantifiers and logical connectives to express the fact that a quadratic polynomial with real number coefficients has at most two real roots.
45. Determine the truth value of the statement $\forall x \exists y (xy = 1)$ if the domain for the variables consists of

- a) the nonzero real numbers.
- b) the nonzero integers.
- c) the positive real numbers.

46. Determine the truth value of the statement $\exists x \forall y (x \leq y^2)$ if the domain for the variables consists of

- a) the positive real numbers.
- b) the integers.
- c) the nonzero real numbers.

47. Show that the two statements $\neg \exists x \forall y P(x, y)$ and $\forall x \exists y \neg P(x, y)$, where both quantifiers over the first variable in $P(x, y)$ have the same domain, and both quantifiers over the second variable in $P(x, y)$ have the same domain, are logically equivalent.

*48. Show that $\forall x P(x) \vee \forall x Q(x)$ and $\forall x \forall y (P(x) \vee Q(y))$, where all quantifiers have the same nonempty domain, are logically equivalent. (The new variable y is used to combine the quantifications correctly.)

*49. a) Show that $\forall x P(x) \wedge \exists x Q(x)$ is logically equivalent to $\forall x \exists y (P(x) \wedge Q(y))$, where all quantifiers have the same nonempty domain.

b) Show that $\forall x P(x) \vee \exists x Q(x)$ is equivalent to $\forall x \exists y (P(x) \vee Q(y))$, where all quantifiers have the same nonempty domain.

A statement is in **prenex normal form (PNF)** if and only if it is of the form

$$Q_1 x_1 Q_2 x_2 \cdots Q_k x_k P(x_1, x_2, \dots, x_k),$$

where each Q_i , $i = 1, 2, \dots, k$, is either the existential quantifier or the universal quantifier, and $P(x_1, \dots, x_k)$ is a predicate involving no quantifiers. For example, $\exists x \forall y (P(x, y) \wedge Q(y))$ is in prenex normal form, whereas $\exists x P(x) \vee \forall x Q(x)$ is not (because the quantifiers do not all occur first).

Every statement formed from propositional variables, predicates, **T**, and **F** using logical connectives and quantifiers is equivalent to a statement in prenex normal form. Exercise 51 asks for a proof of this fact.

*50. Put these statements in prenex normal form. [Hint: Use logical equivalence from Tables 6 and 7 in Section 1.3, Table 2 in Section 1.4, Example 19 in Section 1.4, Exercises 47 and 48 in Section 1.4, and Exercises 48 and 49.]

- a) $\exists x P(x) \vee \exists x Q(x) \vee A$, where A is a proposition not involving any quantifiers
- b) $\neg(\forall x P(x) \vee \forall x Q(x))$
- c) $\exists x P(x) \rightarrow \exists x Q(x)$

**51. Show how to transform an arbitrary statement to a statement in prenex normal form that is equivalent to the given statement. (Note: A formal solution of this exercise requires use of structural induction, covered in Section 5.3.)

*52. Express the quantification $\exists! x P(x)$, introduced in Section 1.4, using universal quantifications, existential quantifications, and logical operators.